

STA304 Midterm SOLUTIONS (Lec0201)

Tuesday October 18, 2022

First Name: _____

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Instructions:

- Total points: 20
- The test is 75 minutes long.
- There are 12 pages including this page.
- Questions begin on the next page.
- Only this exam book will be marked. Write your answers only in the space provided in this booklet. Any work on the Formula Sheet booklet will not be graded.
- Total questions: 12 (10 multiple choice, 2 short answer)
- Please ensure you **fill in your multiple Choice selections (Q1 - Q10) in the scantron on page 12**. Multiple choice answers NOT on the scantron will NOT be graded.
- Please use a calculator and do not round numbers in calculations until the very end, to avoid rounding error.
- In the multiple choice options, please select the closest answer (if the options are numeric) or the most appropriate answer (if the options are text).
- If you believe there to be a typo please still answer the question (see bullet point above), but you can list the question here and we will look into it after:

Multiple Choice Questions

1. d, 2. d, 3. e, 4. a, 5. e, 6. d, 7. a, 8. a, 9. c, 10. a

Q1 (1 point)

What is the sample size required to be 90% confident of the population mean, within 3 units, if we are using simple random sampling on a population of size 500 with population **standard deviation** of 12?

- a. 1
- b. 11
- c. 26
- d. 39
- e. 68

Q2 (1 point)

The following question was asked of a random sample of University of Toronto students: “Do you like statistics and are you interested in pursuing studies in Statistics? Yes or No”.

Which of the following statements are true?

- a. The question is a leading question because it pressures the reader to answer “Yes”.
- b. The question is a loaded question because it pressures the reader to answer “Yes”.
- c. The question is a leading question because it is asking multiple questions.
- d. The question is a loaded question because it is asking multiple questions.
- e. The question is neither a leading question nor a loaded question.

Q3 (1 point)

Assume a Simple Random Sample (without replacement) was conducted on kindergarten children. Assume the study variable of interest is age and the population size was $N=10$. The collected sample was: $\{3, 4, 5, 4\}$.

Calculate the upper bound of the 99% confidence interval for the overall population mean.

- a. 3.18
- b. 4.32
- c. 4.52
- d. 4.57
- e. 4.81

Q4 (1 point)

A hospital has performed cluster sampling by surveying nurses in three randomly selected departments (out of 20 departments, all of equal size). The hospital has 1000 nurses in total, 50 of which are in each of the three selected departments (50 in department A, 50 in department B and 50 in department C). The hospital is interested in estimating the *total* shift time of the nurses. Which ONE of the following lines of code provides appropriate summaries of the different clusters?

a.

```
hospital_data %>%  
  group_by(department) %>%  
  summarize(n=n(), y=sum(shift_length))
```

b.

```
hospital_data %>%  
  filter(department=="A"|department=="B") %>%  
  summarize(n=n(department))
```

c.

```
hospital_data %>%  
  filter(department=="A"|department=="B") %>%  
  select(length(shift_length), mean(shift_length), sd(shift_length))
```

d.

```
hospital_data %>%  
  select(shift_length) %>%  
  group_by(department) %>%  
  summarize(n=n(), y=sum(department), s=sd(department))
```

e.

```
hospital_data %>%  
  select(department) %>%  
  mutate(n=n(), y=mean(shift_length))
```

Q5 (1 point)

A hospital has performed cluster sampling by surveying nurses in three randomly selected departments (out of 20 departments, all of equal size). The hospital has 1000 nurses in total, 50 of which are in each of the three selected departments (50 in department A, 50 in department B, and 50 in department C). The hospital is interested in estimating the *total* shift time of the nurses. Use the following summary table of the `shift_length` variable (from the previous question) to calculate $v(\hat{T}_y)$.

department	n	total_shift_length
A	50	600
B	50	400
C	50	500

- a. 3,825
- b. 25,500
- c. 56,667
- d. 850,000
- e. 1,133,333

Q6 (1 point)

Which ONE of the following is FALSE?

- a. Snowball sampling is a non-probability based sampling approach that is helpful when trying to survey underground populations, such as the homeless.
- b. Convenience sampling is a non-probability based sampling approach that is biased as you are not randomly selecting your observations.
- c. In stratified sampling, we are ensuring that we are selecting some observations from each stratum.
- d. Quota sampling is a probability based sampling approach that randomly selects observations from different clusters in the population.
- e. It is a good idea to test your survey before you implement it to your sample, to help improve your survey design (i.e., reduce grammatical errors, avoid leading questions, etc.).

Q7 (1 point)

A researcher wishes to run a hypothesis test of $H_0 : R = 1$ vs $H_A : R \neq 1$ to make inference on the mass to height ratio of people in their study. Assume this was a simple random sample of $n = 87$, from a population is of size $N = 500$. What is the test statistic of interest based on the output below.

```
my.design <- svydesign(id=~1, data=starwars, fpc=rep(500, 87))

svyratio(~mass, ~height, design=my.design, na.rm=TRUE)

## Ratio estimator: svyratio.survey.design2(~mass, ~height, design = my.design, na.rm = TRUE)
## Ratios=
##      height
## mass 0.5592091
## SEs=
##      height
## mass 0.1138844
```

- a. -3.87
- b. 4.91
- c. 43.12
- d. -0.41
- e. 3.64

Q8 (1 point)

A company of $N = 20$ employees wishes to take a random sample of 5 different employees. To do this they write out the list of employee names in alphabetical order, randomly pick a starting name and then select every fourth employee on the list. What is the probability of any one sample of 5 employees being collected?

- a. 0.25000
- b. 0.20000
- c. 0.01250
- d. 0.00021
- e. 0.00006

Q9 (1 point)

We wish to estimate the following model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_{\text{hours}} + \beta_2 x_{\text{minor}} + \beta_3 x_{\text{specialist}}$$

Note the data has the following variables:

- `passed_course` is a categorical variable with 2 categories (1 if they passed, and 0 if not).
- `program_of_study` is a categorical variable with 3 categories (Major, Minor, Specialist).
- `hours_slept` is a numeric variable.

Which ONE of the following would be best to run in order to estimate the model above?

a.

```
my_model <- glm(passed_course ~ hours_slept + program_of_study)
summary(my_model)
```

b.

```
my_model <- lm(passed_course ~ hours_slept + program_of_study, family="binomial")
summary(my_model)
```

c.

```
my_model <- glm(passed_course ~ hours_slept + program_of_study, family="binomial")
summary(my_model)
```

d.

```
my_model <- brm(passed_course ~ hours_slept + program_of_study)
summary(my_model)
```

- e. None of the above chunks are appropriate, the model is incomplete because it is missing an error term (ϵ).

Q10 (1 point)

Which ONE of the following best describes the difference between a Bayesian and Frequentist approach?

- a. A Bayesian approach invokes a prior distribution on the parameter of interest and then updates the distribution based on collected data.
- b. There is no difference
- c. It is always best to use a Frequentist approach.
- d. It is always best to use a Bayesian approach.
- e. A Frequentist approach invokes a prior distribution on the parameter of interest and then updates the distribution based on collected data.

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STA304-Fall2022
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Short Answer Question

Q11 (5 points total)

A grocery store is interested in learning about the number of overtime hours used by its employees. The store has 100 employees in 5 different departments (all of equal size) and can only afford to survey 40 employees.

- a. (2 points) The grocery store is not sure if they should treat departments as clusters or strata. Explain in 2 or 3 sentences what advice would you give the grocery store to help them decide between stratified and cluster sampling.

Note: Please ensure you demonstrate your understanding of the difference between stratified and cluster sampling.

If the departments are expected to be similar then the departments should be treated as clusters. (1 point)

If the departments are expected to be different then the departments should be treated as strata. (1 point)

Short Answer Question

Q11 (5 points total)

- b. (3 points) If the grocery decided to treat departments as *strata* describe how they could use the `sample()` function to select their sample. Note: if you feel more comfortable you could write a line of code here and explain it.

Answers may vary

- 1 point for correct demonstration of inputs/arguments in the `sample()` function
- 1 point for correct understanding of outputs of the `sample()` function.
- 1 point for correct understanding of cluster sampling.

Example:

The company could use the following code 5 times (once for each department) to randomly select the 8 employees to be in the sample, from each department.

```
sample(c("Employee1","Employee2",...,"Employee20"), size=8, replace=FALSE)
```

Short Answer Question

Q12 (5 points total)

Assume you have a sample of $y_1, y_2, \dots, y_n$ taken from the population of $\{u_1, u_2, \dots, u_N\}$. Recall, $E(Y_i) = \mu$, $Var(Y_i) = \sigma^2$.

- a. (1 point) Show that $E(Y_i) = \mu$. I.e., Show that the expected value, of a sample of size $n = 1$, is the population mean.

(1 point)

$$E(Y_i) = \sum_{i=1}^N u_i \times \frac{1}{N} = \mu$$

Short Answer Question

Q12 (5 points total)

- b. (4 points) Derive $E[Y_i(1 - Y_i)]$. Write it as an expression of μ and σ^2 .

Recall: $Var(Y_i) = E[(Y_i - E(Y_i))^2]$

Let's start by expanding out the hint:

$$Var(Y_i) = E[(Y_i - E(Y_i))^2]$$

$$Var(Y_i) = E[(Y_i - \mu)^2]$$

$$Var(Y_i) = E[Y_i^2 - 2Y_i\mu + \mu^2]$$

$$Var(Y_i) = E[Y_i^2] - 2\mu \times E[Y_i] + E[\mu^2]$$

$$Var(Y_i) = E[Y_i^2] - 2\mu \times \mu + \mu^2$$

$$Var(Y_i) = E[Y_i^2] - 2\mu^2 + \mu^2$$

$$Var(Y_i) = E[Y_i^2] - \mu^2$$

$$E[Y_i^2] = Var(Y_i) + \mu^2$$

$$E[Y_i^2] = \sigma^2 + \mu^2$$

$$E(Y_i(1 - Y_i)) = E(Y_i - Y_i^2) = E(Y_i) - E(Y_i^2) = \mu - (\sigma^2 + \mu^2)$$